

Generation Interconnection System Impact Study Report

For Queue Project Q18-1010

Acec Cottonville 100 kV
May 17, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1010. The Interconnection Customer proposes a 172.5 MW solar photovoltaic facility interconnecting at the Acec Cottonville 100 kV substation in the Indiana area. The request was received on June 05, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 286 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$929,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Ligonier - Montibello 100 KV, the Bear Garden - Ward Gulf 100 KV loads to 126.7% of its 109 MVA rating. Project DFAX: 31.9%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ward Gulf - Bear Garen 100 KV Line	Rebuild line with higher-capacity conductor	\$2,170,000	43% (\$929,000)
	Total allocated to project		\$929,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.